



1. Description

1.1. Project

Project Name	Phenix2025-407vgt6
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	12/17/2025

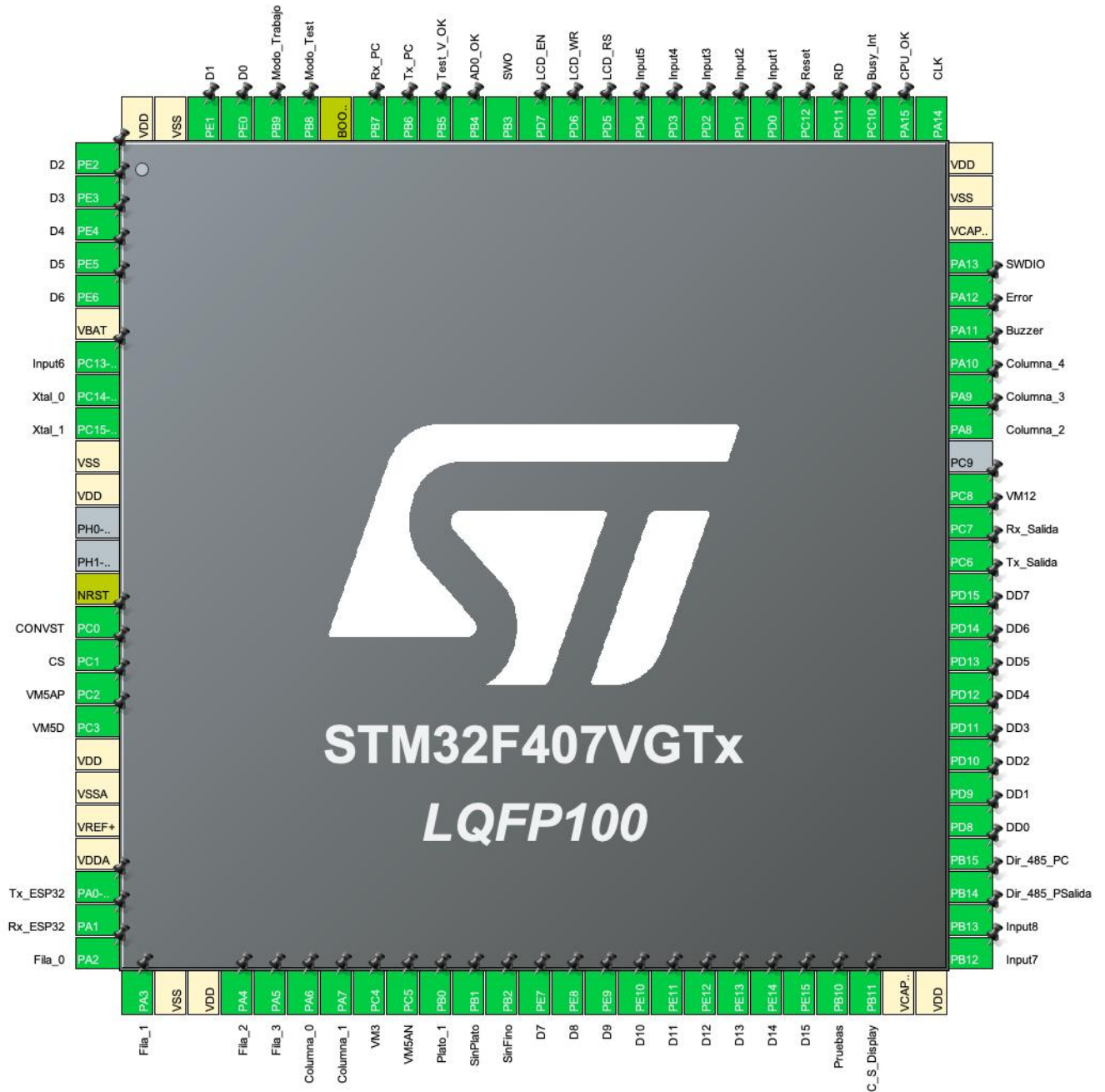
1.2. MCU

MCU Series	STM32F4
MCU Line	STM32F407/417
MCU name	STM32F407VGTx
MCU Package	LQFP100
MCU Pin number	100

1.3. Core(s) information

Core(s)	Arm Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

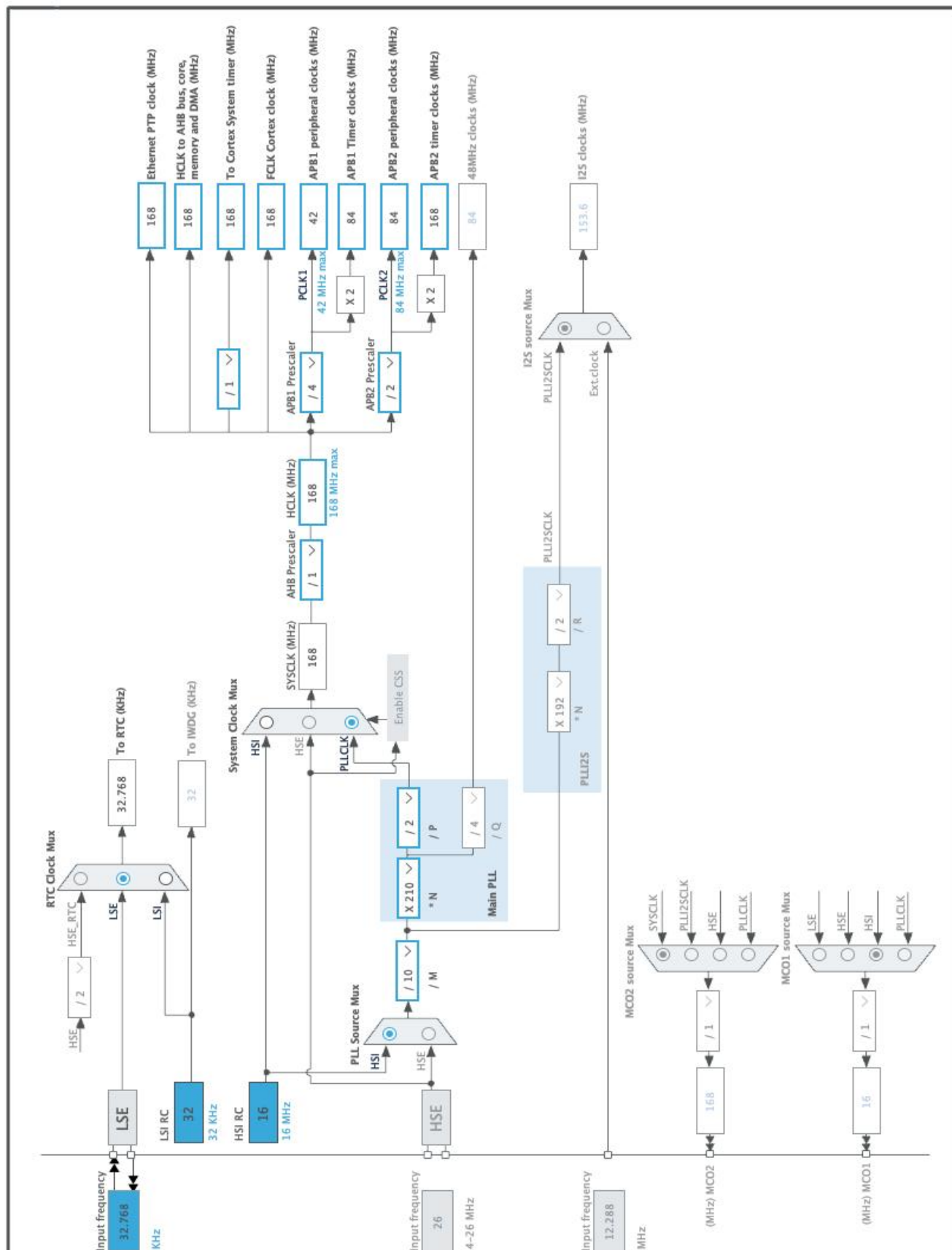
Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2 *	I/O	GPIO_Input	D2
2	PE3 *	I/O	GPIO_Input	D3
3	PE4 *	I/O	GPIO_Input	D4
4	PE5 *	I/O	GPIO_Input	D5
5	PE6 *	I/O	GPIO_Input	D6
6	VBAT	Power		
7	PC13-ANTI_TAMP *	I/O	GPIO_Input	Input6
8	PC14-OSC32_IN	I/O	RCC_OSC32_IN	Xtal_0
9	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	Xtal_1
10	VSS	Power		
11	VDD	Power		
14	NRST	Reset		
15	PC0 *	I/O	GPIO_Output	CONVST
16	PC1 *	I/O	GPIO_Output	CS
17	PC2	I/O	ADC1_IN12	VM5AP
18	PC3	I/O	ADC1_IN13	VM5D
19	VDD	Power		
20	VSSA	Power		
21	VREF+	Power		
22	VDDA	Power		
23	PA0-WKUP	I/O	UART4_TX	Tx_ESP32
24	PA1	I/O	UART4_RX	Rx_ESP32
25	PA2 *	I/O	GPIO_Output	Fila_0
26	PA3 *	I/O	GPIO_Output	Fila_1
27	VSS	Power		
28	VDD	Power		
29	PA4 *	I/O	GPIO_Output	Fila_2
30	PA5 *	I/O	GPIO_Output	Fila_3
31	PA6 *	I/O	GPIO_Input	Columna_0
32	PA7 *	I/O	GPIO_Input	Columna_1
33	PC4	I/O	ADC1_IN14	VM3
34	PC5	I/O	ADC1_IN15	VM5AN
35	PB0	I/O	GPIO_EXTI0	Plato_1
36	PB1	I/O	GPIO_EXTI1	SinPlato
37	PB2	I/O	GPIO_EXTI2	SinFino
38	PE7 *	I/O	GPIO_Input	D7

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
39	PE8 *	I/O	GPIO_Input	D8
40	PE9 *	I/O	GPIO_Input	D9
41	PE10 *	I/O	GPIO_Input	D10
42	PE11 *	I/O	GPIO_Input	D11
43	PE12 *	I/O	GPIO_Input	D12
44	PE13 *	I/O	GPIO_Input	D13
45	PE14 *	I/O	GPIO_Input	D14
46	PE15 *	I/O	GPIO_Input	D15
47	PB10 *	I/O	GPIO_Input	Pruebas
48	PB11 *	I/O	GPIO_Input	C_S_Display
49	VCAP_1	Power		
50	VDD	Power		
51	PB12 *	I/O	GPIO_Input	Input7
52	PB13 *	I/O	GPIO_Input	Input8
53	PB14 *	I/O	GPIO_Output	Dir_485_PSalida
54	PB15 *	I/O	GPIO_Output	Dir_485_PC
55	PD8 *	I/O	GPIO_Output	DD0
56	PD9 *	I/O	GPIO_Output	DD1
57	PD10 *	I/O	GPIO_Output	DD2
58	PD11 *	I/O	GPIO_Output	DD3
59	PD12 *	I/O	GPIO_Output	DD4
60	PD13 *	I/O	GPIO_Output	DD5
61	PD14 *	I/O	GPIO_Output	DD6
62	PD15 *	I/O	GPIO_Output	DD7
63	PC6	I/O	USART6_TX	Tx_Salida
64	PC7	I/O	USART6_RX	Rx_Salida
65	PC8 *	I/O	GPIO_Input	VM12
67	PA8 *	I/O	GPIO_Input	Columna_2
68	PA9 *	I/O	GPIO_Input	Columna_3
69	PA10 *	I/O	GPIO_Input	Columna_4
70	PA11 *	I/O	GPIO_Output	Buzzer
71	PA12 *	I/O	GPIO_Output	Error
72	PA13	I/O	SYS_JTMS-SWDIO	SWDIO
73	VCAP_2	Power		
74	VSS	Power		
75	VDD	Power		
76	PA14	I/O	SYS_JTCK-SWCLK	CLK
77	PA15 *	I/O	GPIO_Output	CPU_OK
78	PC10	I/O	GPIO_EXTI10	Busy_Int

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
79	PC11 *	I/O	GPIO_Output	RD
80	PC12 *	I/O	GPIO_Output	Reset
81	PD0 *	I/O	GPIO_Input	Input1
82	PD1 *	I/O	GPIO_Input	Input2
83	PD2 *	I/O	GPIO_Input	Input3
84	PD3 *	I/O	GPIO_Input	Input4
85	PD4 *	I/O	GPIO_Input	Input5
86	PD5 *	I/O	GPIO_Output	LCD_RS
87	PD6 *	I/O	GPIO_Output	LCD_WR
88	PD7 *	I/O	GPIO_Output	LCD_EN
89	PB3	I/O	SYS_JTDO-SWO	SWO
90	PB4 *	I/O	GPIO_Output	AD0_OK
91	PB5 *	I/O	GPIO_Output	Test_V_OK
92	PB6	I/O	USART1_TX	Tx_PC
93	PB7	I/O	USART1_RX	Rx_PC
94	BOOT0	Boot		
95	PB8 *	I/O	GPIO_Output	Modo_Test
96	PB9 *	I/O	GPIO_Output	Modo_Trabajo
97	PE0 *	I/O	GPIO_Input	D0
98	PE1 *	I/O	GPIO_Input	D1
99	VSS	Power		
100	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F4
Line	STM32F407/417
MCU	STM32F407VGTx
Datasheet	DS8626 Rev8

1.2. Parameter Selection

Temperature	25
Vdd	3.3

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

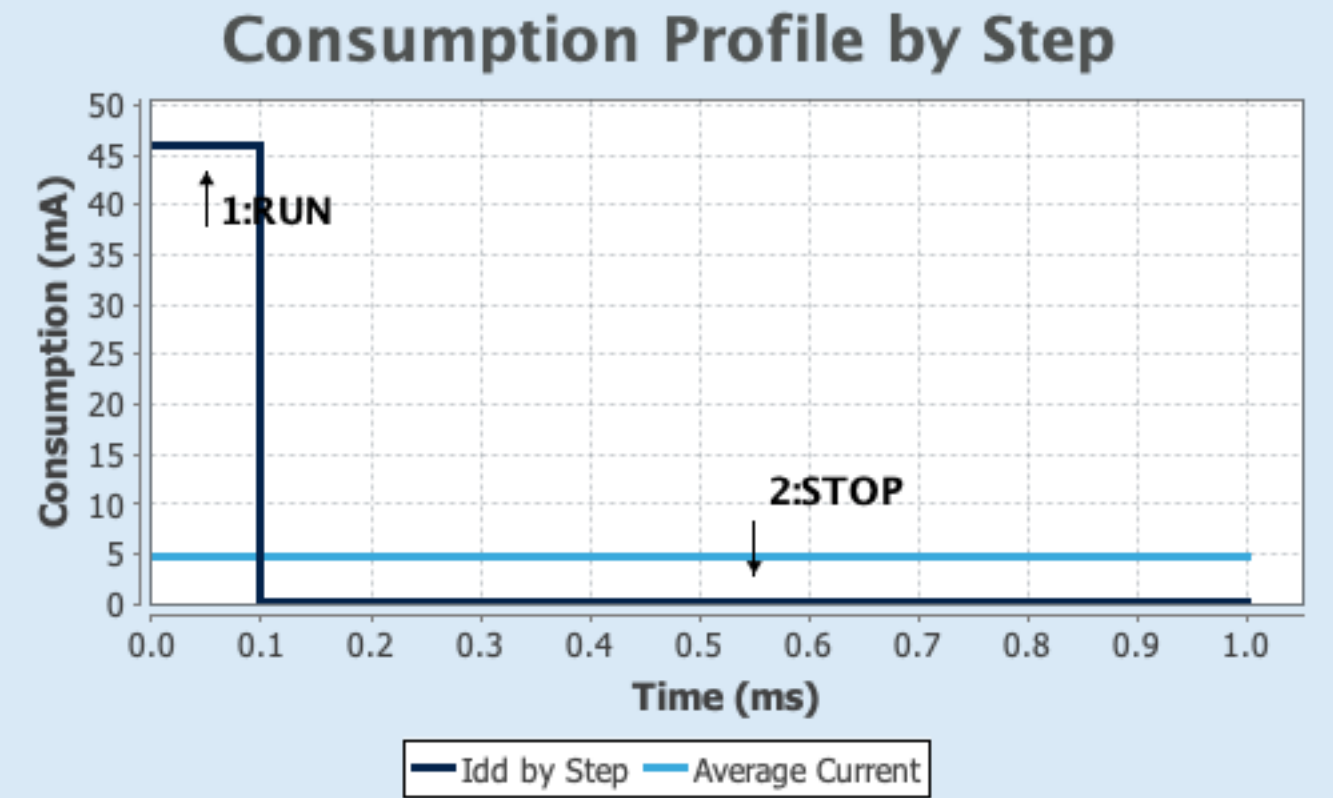
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.3	3.3
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	FLASH	n/a
CPU Frequency	168 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LP Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	46 mA	280 μ A
Duration	0.1 ms	0.9 ms
DMIPS	210.0	0.0
Ta Max	98.47	104.96
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	4.85 mA
Battery Life	29 days, 4 hours	Average DMIPS	210.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	Phenix2025-407vgt6
Project Folder	/Users/carlos/Trabajo/ST/WorkSpace/Phenix2025-407vgt6
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_F4 V1.28.2
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	Yes
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_ADC1_Init	ADC1
5	MX_RTC_Init	RTC
6	MX_TIM6_Init	TIM6
7	MX_UART4_Init	UART4
8	MX_USART6_UART_Init	USART6
9	MX_USART1_UART_Init	USART1

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: IN12

mode: IN13

mode: IN14

mode: IN15

mode: Vbat Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler PCLK2 divided by 4

Resolution 12 bits (15 ADC Clock cycles)

Data Alignment Right alignment

Scan Conversion Mode Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

End Of Conversion Selection EOC flag at the end of single channel conversion

ADC_Regular_ConversionMode:

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 12

Sampling Time 3 Cycles

ADC_Injected_ConversionMode:

Number Of Conversions 0

WatchDog:

Enable Analog WatchDog Mode false

3.2. RCC

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.2.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	5 WS (6 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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3.3. RTC

mode: Activate Clock Source

mode: Activate Calendar

3.3.1. Parameter Settings:

General:

Hour Format	Hourformat 24
Asynchronous Predivider value	127
Synchronous Predivider value	255

Calendar Time:

Data Format	BCD data format
Hours	12 *
Minutes	0
Seconds	0
Day Light Saving: value of hour adjustment	Daylightsaving None
Store Operation	Storeoperation Reset

Calendar Date:

Week Day	Wednesday *
Month	December *
Date	1
Year	25 *

3.4. SYS

Debug: Trace Asynchronous Sw

Timebase Source: SysTick

3.5. TIM6

mode: Activated

3.5.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	840 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	99 *
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)
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3.6. UART4

Mode: Asynchronous

3.6.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

3.7. USART1

Mode: Asynchronous

3.7.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

3.8. USART6

Mode: Asynchronous

3.8.1. Parameter Settings:

Basic Parameters:

Baud Rate	9600 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC2	ADC1_IN12	Analog mode	No pull-up and no pull-down	n/a	VM5AP
	PC3	ADC1_IN13	Analog mode	No pull-up and no pull-down	n/a	VM5D
	PC4	ADC1_IN14	Analog mode	No pull-up and no pull-down	n/a	VM3
	PC5	ADC1_IN15	Analog mode	No pull-up and no pull-down	n/a	VM5AN
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	Xtal_0
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	Xtal_1
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	SWDIO
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	CLK
	PB3	SYS_JTDO-SWO	n/a	n/a	n/a	SWO
UART4	PA0-WKUP	UART4_TX	Alternate Function Push Pull	Pull-up *	Very High *	Tx_ESP32
	PA1	UART4_RX	Alternate Function Push Pull	Pull-up *	Very High *	Rx_ESP32
USART1	PB6	USART1_TX	Alternate Function Push Pull	Pull-up *	Very High *	Tx_PC
	PB7	USART1_RX	Alternate Function Push Pull	Pull-up *	Very High *	Rx_PC
USART6	PC6	USART6_TX	Alternate Function Push Pull	Pull-up *	Very High *	Tx_Salida
	PC7	USART6_RX	Alternate Function Push Pull	Pull-up *	Very High *	Rx_Salida
GPIO	PE2	GPIO_Input	Input mode	Pull-up *	n/a	D2
	PE3	GPIO_Input	Input mode	Pull-up *	n/a	D3
	PE4	GPIO_Input	Input mode	Pull-up *	n/a	D4
	PE5	GPIO_Input	Input mode	Pull-up *	n/a	D5
	PE6	GPIO_Input	Input mode	Pull-up *	n/a	D6
	PC13-ANTI_TAMP	GPIO_Input	Input mode	Pull-up *	n/a	Input6
	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	High *	CONVST

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	High *	CS
	PA2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	Fila_0
	PA3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	Fila_1
	PA4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	Fila_2
	PA5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	Fila_3
	PA6	GPIO_Input	Input mode	Pull-up *	n/a	Columna_0
	PA7	GPIO_Input	Input mode	Pull-up *	n/a	Columna_1
	PB0	GPIO_EXTI0	External Interrupt Mode with Falling edge trigger detection	Pull-up *	n/a	Plato_1
	PB1	GPIO_EXTI1	External Interrupt Mode with Falling edge trigger detection	Pull-up *	n/a	SinPlato
	PB2	GPIO_EXTI2	External Interrupt Mode with Falling edge trigger detection	Pull-up *	n/a	SinFino
	PE7	GPIO_Input	Input mode	Pull-up *	n/a	D7
	PE8	GPIO_Input	Input mode	Pull-up *	n/a	D8
	PE9	GPIO_Input	Input mode	Pull-up *	n/a	D9
	PE10	GPIO_Input	Input mode	Pull-up *	n/a	D10
	PE11	GPIO_Input	Input mode	Pull-up *	n/a	D11
	PE12	GPIO_Input	Input mode	Pull-up *	n/a	D12
	PE13	GPIO_Input	Input mode	Pull-up *	n/a	D13
	PE14	GPIO_Input	Input mode	Pull-up *	n/a	D14
	PE15	GPIO_Input	Input mode	Pull-up *	n/a	D15
	PB10	GPIO_Input	Input mode	Pull-up *	n/a	Pruebas
	PB11	GPIO_Input	Input mode	Pull-up *	n/a	C_S_Display
	PB12	GPIO_Input	Input mode	Pull-up *	n/a	Input7
	PB13	GPIO_Input	Input mode	Pull-up *	n/a	Input8
	PB14	GPIO_Output	Output Push Pull	Pull-down *	High *	Dir_485_PSalida
	PB15	GPIO_Output	Output Push Pull	Pull-down *	High *	Dir_485_PC
	PD8	GPIO_Output	Output Push Pull	Pull-up *	Medium *	DD0
	PD9	GPIO_Output	Output Push Pull	Pull-up *	Medium *	DD1
	PD10	GPIO_Output	Output Push Pull	Pull-up *	Medium *	DD2
	PD11	GPIO_Output	Output Push Pull	Pull-up *	Medium *	DD3

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PD12	GPIO_Output	Output Push Pull	Pull-up *	Medium *	DD4
	PD13	GPIO_Output	Output Push Pull	Pull-up *	Medium *	DD5
	PD14	GPIO_Output	Output Push Pull	Pull-up *	Medium *	DD6
	PD15	GPIO_Output	Output Push Pull	Pull-up *	Medium *	DD7
	PC8	GPIO_Input	Input mode	Pull-up *	n/a	VM12
	PA8	GPIO_Input	Input mode	Pull-up *	n/a	Columna_2
	PA9	GPIO_Input	Input mode	Pull-up *	n/a	Columna_3
	PA10	GPIO_Input	Input mode	Pull-up *	n/a	Columna_4
	PA11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	Buzzer
	PA12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	Error
	PA15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	CPU_OK
	PC10	GPIO_EXTI10	External Interrupt Mode with Falling edge trigger detection	Pull-up *	n/a	Busy_Int
	PC11	GPIO_Output	Output Push Pull	Pull-up *	High *	RD
	PC12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	Reset
	PD0	GPIO_Input	Input mode	Pull-up *	n/a	Input1
	PD1	GPIO_Input	Input mode	Pull-up *	n/a	Input2
	PD2	GPIO_Input	Input mode	Pull-up *	n/a	Input3
	PD3	GPIO_Input	Input mode	Pull-up *	n/a	Input4
	PD4	GPIO_Input	Input mode	Pull-up *	n/a	Input5
	PD5	GPIO_Output	Output Push Pull	Pull-up *	Medium *	LCD_RS
	PD6	GPIO_Output	Output Push Pull	Pull-up *	Medium *	LCD_WR
	PD7	GPIO_Output	Output Push Pull	Pull-up *	Medium *	LCD_EN
	PB4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	AD0_OK
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	Test_V_OK
	PB8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	Modo_Test
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	Modo_Trabajo
	PE0	GPIO_Input	Input mode	Pull-up *	n/a	D0
	PE1	GPIO_Input	Input mode	Pull-up *	n/a	D1

4.2. DMA configuration

DMA request	Stream	Direction	Priority
UART4_TX	DMA1_Stream4	Memory To Peripheral	High *
USART1_TX	DMA2_Stream7	Memory To Peripheral	High *
USART6_TX	DMA2_Stream6	Memory To Peripheral	High *
UART4_RX	DMA1_Stream2	Peripheral To Memory	Medium *
USART1_RX	DMA2_Stream2	Peripheral To Memory	Medium *

UART4_TX: DMA1_Stream4 DMA request Settings:

Mode: Normal
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

USART1_TX: DMA2_Stream7 DMA request Settings:

Mode: Normal
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

USART6_TX: DMA2_Stream6 DMA request Settings:

Mode: Normal
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

UART4_RX: DMA1_Stream2 DMA request Settings:

Mode: Normal
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

USART1_RX: DMA2_Stream2 DMA request Settings:

Mode: Normal
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
EXTI line0 interrupt	true	0	0
EXTI line1 interrupt	true	0	0
EXTI line2 interrupt	true	0	0
DMA1 stream2 global interrupt	true	2	0
DMA1 stream4 global interrupt	true	2	0
ADC1, ADC2 and ADC3 global interrupts	true	4	0
USART1 global interrupt	true	2	0
EXTI line[15:10] interrupts	true	0	0
UART4 global interrupt	true	2	0
TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	true	1	0
DMA2 stream2 global interrupt	true	2	0
DMA2 stream6 global interrupt	true	2	0
DMA2 stream7 global interrupt	true	2	0
USART6 global interrupt	true	2	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	false	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	false	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
System service call via SWI instruction	false	false	false
Debug monitor	false	false	false
Pendable request for system service	false	false	false
System tick timer	false	true	true
EXTI line0 interrupt	true	true	true
EXTI line1 interrupt	true	true	true
EXTI line2 interrupt	true	true	true
DMA1 stream2 global interrupt	true	true	true
DMA1 stream4 global interrupt	true	true	true
ADC1, ADC2 and ADC3 global interrupts	true	true	true
USART1 global interrupt	true	true	true
EXTI line[15:10] interrupts	true	true	true
UART4 global interrupt	true	true	true
TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	true	true	true
DMA2 stream2 global interrupt	true	true	true
DMA2 stream6 global interrupt	true	true	true
DMA2 stream7 global interrupt	true	true	true
USART6 global interrupt	true	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32f405-415_407-417_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32f405-415_407-417_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32f4-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

- Application Notes https://www.st.com/resource/en/application_note/an2945-stm8s-and-stm32-mcus-a-consistent-832bit-product-line-for-painless-migration-stmicroelectronics.pdf
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